An Orthogonal Semilattice Action Whose Universal Crossed Product Is Not the ℓ^1 Quotient

Candidate counterexample packet
arXiv:2601.14907, Remark 27

12 August 2026

Status. Candidate full counterexample to the *stronger equality speculation* in Remark 27, subject to expert review. The construction does not answer the remark's preceding question about faithfulness of the maximal seminorm modulo $\mathcal{N}$; in this example that seminorm is faithful.

1 Source question and exact scope

Bardadyn–Kwaśniewski define the universal Banach crossed product $A \rtimes_\alpha S$ as the Hausdorff completion of $\ell^1(\alpha)$ for the maximal seminorm over covariant representations. Their closed ideal $\mathcal{N}$ is generated by the order relations $a\delta_s - a\delta_t$ for $a \in I_s$ and $s \le t$. Remark 27 on page 13 reads:

> "In fact, it is often the case (though we do not know in general) that the seminorm $\|\cdot\|_{\max}$ on $\ell^1(\alpha)$ becomes a norm on $\ell^1(\alpha)/\mathcal{N}$. It might even be that $A \rtimes_\alpha S = \ell^1(\alpha)/\mathcal{N}$."

The theorem below refutes exactly the second sentence. It shows that the canonical dense map from the Banach quotient into the crossed product need not be onto and that the two natural norms can be arbitrarily far apart.

Remark 27. As we noted above, instead of completions of $\ell^1(\alpha)$ one could consider completions of the quotient algebra $\ell^1(\alpha)/\mathcal{N}$. In fact, it is often the case (though we do not know in general) that the seminorm $\|\cdot\|_{\max}$ on $\ell^1(\alpha)$ becomes a norm on $\ell^1(\alpha)/\mathcal{N}$. It might even be that $A \rtimes_\alpha S = \ell^1(\alpha)/\mathcal{N}$.

Figure 1: Remark 27 in the source PDF, page 13.

2 Definitions

Let

$$S = \{0, e_1, e_2, \ldots\}, \qquad e_n^2 = e_n, \quad e_n e_m = 0 \ (n \neq m), \quad 0s = s0 = 0.$$

This is a commutative inverse semigroup: every element is its own inverse. Let $A = c_0(\mathbb{N})$ with coordinatewise multiplication, write p_n for the nth coordinate vector, and set

$$I_0 = \{0\}, \qquad I_{e_n} = \mathbb{C}p_n, \qquad \alpha_s = \mathrm{id}_{I_s}.$$

For this action, $\ell^1(\alpha)$ consists of sections $f(e_n) = a_n p_n$ with $\sum_n |a_n| < \infty$.

3 Proof Intuition

The semigroup fibers are mutually orthogonal. Consequently convolution has no cross terms and the order-relation ideal is zero, so the source quotient retains the ℓ^1 norm. Covariance has the opposite effect on every integrated representation: (CR3) erases the fiber labels and turns a section into its coefficient sum inside c_0. The identity representation of c_0 attains the resulting sup norm. Thus the universal construction completes the same algebraic sequences from ℓ^1 to c_0.

Theorem 1. *The family above is an inverse-semigroup action in the sense of [1]. There are canonical isometric identifications*

$$\ell^1(\alpha)/\mathcal{N} \cong (\ell^1, \|\cdot\|_1), \qquad A \rtimes_\alpha S \cong (c_0, \|\cdot\|_\infty),$$

under which the canonical map is the usual dense inclusion $\ell^1 \hookrightarrow c_0$. In particular

$$A \rtimes_\alpha S \neq \ell^1(\alpha)/\mathcal{N}$$

canonically, even as Banach algebras, and the quotient norm is not equivalent to the maximal norm. The maximal seminorm nevertheless has kernel exactly $\mathcal{N} = 0$ in this example.

Proof. For idempotents e_n, e_m, the composition of the two partial identity maps is the identity on $I_{e_n} \cap I_{e_m} = I_{e_n e_m}$. Thus $s \mapsto \alpha_s$ is a semigroup homomorphism into the partial automorphisms of A. Every nonzero fiber ideal has the norm-one unit p_n, $I_0 = 0$, and $\operatorname{span} \bigcup_s I_s = c_{00}$ is dense in c_0. Hence this is an action satisfying the source hypotheses.

Identify a section f with the scalar sequence (a_n) determined by $f(e_n) = a_n p_n$. In the convolution formula, the only pair whose product is e_k is (e_k, e_k). Therefore

$$(f * g)(e_k) = a_k b_k p_k,$$

and all products belonging to the zero fiber vanish. Hence $\ell^1(\alpha)$ is isometrically ℓ^1 with pointwise multiplication.

The natural order on this semilattice is $s \leq t$ iff $s = st$. Its only strict comparisons are $0 \leq t$. In every order generator $a\delta_s - a\delta_t$ with $s < t$, one has $a \in I_0 = 0$; when $s = t$ the difference is zero. Consequently $\mathcal{N} = 0$.

Let (π, v) be any covariant representation of the action in a Banach algebra B. Condition (CR3) gives

$$\pi(a_n p_n) v_{e_n} = \pi(a_n p_n).$$

For $x = (a_n) \in \ell^1 \subset c_0$, absolute convergence and continuity of the contractive representation π now yield

$$(\pi \times v)(f) = \sum_n \pi(a_n p_n) v_{e_n} = \sum_n \pi(a_n p_n) = \pi(x).$$

It follows that $\|f\|_{\max} \leq \|x\|_\infty$.

For the reverse inequality, take $B = A$, $\pi = \mathrm{id}_A$, $v_0 = 0$, and $v_{e_n} = p_n$ (viewed in $A \subset A''$). If the product of two semigroup elements is zero, (CR2) is vacuous because its coefficient lies in I_0; in the remaining case it says $ap_n^2 = ap_n$. Conditions (CR1) and (CR3) likewise reduce to $p_n a = a p_n = a$ for $a \in I_{e_n}$. Thus this is a covariant representation, and its integrated form sends f to x. Hence

$$\|f\|_{\max} = \|x\|_\infty. \tag{1}$$

The Hausdorff completion of ℓ^1 in the norm (1) is c_0, with the same pointwise product. On the other hand, the quotient by $\mathcal{N} = 0$ retains the complete ℓ^1 norm. The canonical inclusion is not onto: for example, $(1/n)_{n \geq 1}$ lies in $c_0 \setminus \ell^1$. Moreover, for $x^{(N)} = (1, \ldots, 1, 0, \ldots)$,

$$\|x^{(N)}\|_1 = N, \qquad \|x^{(N)}\|_{\max} = 1,$$

so the two norms are not equivalent. This proves every assertion. □

4 Verification

The accompanying script checks finite truncations of the convolution law, the identity covariant representation, and the norm ratio. It also displays that the truncations of $(1/n)$ are Cauchy in sup norm while their ℓ^1 norms diverge. These computations are sanity checks; the proof above is exact and does not depend on them.

The proof uses only (PA1)–(PA2), (CR1)–(CR3), the convolution formula, and the definition of $\mathcal{N}$ from the source. In particular, it does not assume the later disintegration theorem. A reviewer should check the source's natural-order convention and the identity pair against (CR1)–(CR3); both checks are written out above.

5 Limitations and novelty check

This is a full negative answer only to the stronger proposed equality. It does *not* refute the possibility that $\|\cdot\|_{\max}$ is always faithful on $\ell^1(\alpha)/\mathcal{N}$; indeed it is faithful here. Eight focused attempts on that kernel question are recorded in the run attempt note, with the unresolved obstruction isolated as an infinite Banach-colimit/gluing problem.

A bounded search through 12 August 2026 checked the run registry and solution, attempt, and proof-gap indexes; the exact title and wording of Remark 27; the arXiv record and author/title searches; and combinations of “inverse semigroup”, “Banach algebra crossed product”, “maximal seminorm”, and “quotient”. It found the January 2026 source and adjacent C^*-algebraic literature, but no later resolution or this orthogonal-semilattice example. This is a bounded novelty check, not a claim of priority.

Human-review recommendation. Promote after verifying the one-line natural-order calculation and the use of (CR3). The scope label should remain explicit: counterexample to the equality speculation; general faithfulness still open.

References

[1] K. Bardadyn and B. K. Kwaśniewski, *Banach algebra crossed products by inverse semigroup actions*, arXiv:2601.14907, 2026.